

Erdős Problem #708

EP-708

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $g(n)$ be minimal such that for any $A \subseteq [2, \infty) \cap \mathbb{N}$ with $|A| = n$ and any set I of $\max(A)$ consecutive integers there exists some $B \subseteq I$ with $|B| = g(n)$ such that $\prod_{a \in A} a \mid \prod_{b \in B} b$. Is it true that $g(n) \leq (2 + o(1))n$? Or perhaps even $g(n) \leq 2n$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>